

Quantitative Spatial Economics for Policy Evaluation

Topic 1: Rosen–Roback and the quantitative Glaeser–Gottlieb framework

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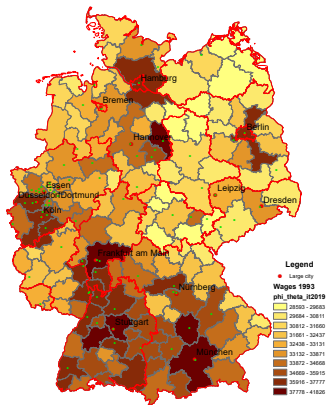
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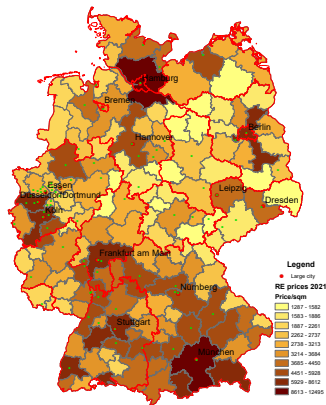
Acknowledgements

- ▶ This slide deck
 - ▶ builds on the lecture "spatial equilibrium across cities" by Tuukka Saarimaa (2021)
 - ▶ builds on the lecture "Rosen-Roback Framework" by Felipe Carozzi (2023)
 - ▶ uses material from Brueckner's textbook Chapter 11

Introduction



(a) Wage 2019



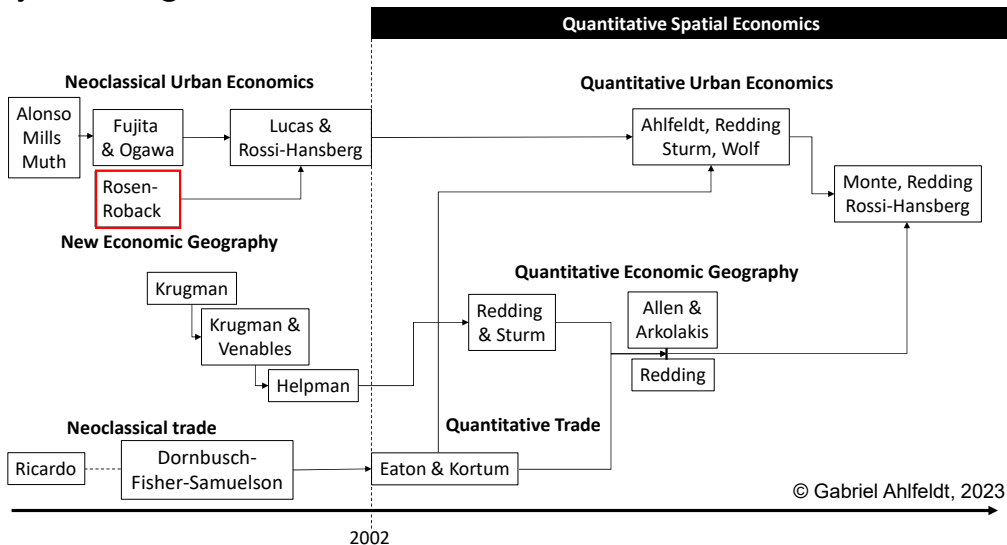
(b) Price/sqm 2021

If one city is an **attractive place to live**, will it have high or low real wages?

Roadmap

- ▶ We analyze the **tradeoff** between income, amenities, and housing costs
 - ▶ across cities or regions
- ▶ We introduce the **spatial equilibrium** assumption
 - ▶ Identical utility levels across regions
 - ▶ Identical firm profits across regions
- ▶ **Rosen-Roback framework** is a core concept in urban economics
 - ▶ Name refers to Rosen (1979) and Roback (1982)
 - ▶ Focus on **Glaeser & Gottlieb (2009)** framework
 - ▶ Introduce mapping from **primitives and endogenous outcomes**
 - ▶ Use mapping for quantification (**estimation and inversion**)
- ▶ Workhorse tool for valuing non-marketed goods

History of thought



Roback (1982)

- ▶ A famous paper by Jennifer Roback's 1980 PhD thesis
 - ▶ over 5200 Google Scholar citations (04/2026)
- ▶ **Not only important to urban economics**, but also to
 - ▶ Labor economics (local labor markets)
 - ▶ Environmental economics (pricing of environmental amenities)
 - ▶ Trade (markets and industry concentration)
 - ▶ Development economics (migration)
 - ▶ Local public finance

Basic assumptions

- ▶ The basic framework for the analysis is a simple **general equilibrium model**
 - ▶ capital and labor are assumed **completely mobile** across cities.
 - ▶ Complete mobility of labor means that the costs of changing residences are zero
 - ▶ Workers are homogenous, **no idiosyncratic tastes for locations**
 - ▶ **Intercity commuting costs** are assumed **prohibitive**
 - ▶ Workers live and work in the same local labour market
 - ▶ **Intracity commuting costs** are **zero**
 - ▶ focus attention on the across-city allocation of workers and firms
 - ▶ **Floor space supply is fixed** within cities
 - ▶ but is assumed interchangeable between uses within a city

Amenities

- ▶ **Consumer utility** depended on the consumption of
 - ▶ a composite commodity c
 - ▶ A basket of freely tradable goods
 - ▶ and housing q
- ▶ Consumers or workers also get utility from **urban amenities**
 - ▶ Different cities have different **exogenous amenities**
 - ▶ E.g., pleasant climate, nature, clean air, low crime
 - ▶ Amenities may be endogenous and depend on city's population
- ▶ **Quality of life** is an index of many different amenities (or disamenities)
 - ▶ we will denote it by a
 - ▶ akin to TFP shifter in production function

Utility function and dependence

- ▶ **Utility** depends on consumption of marketed (c, q) and non-marketed (a) goods

$$u = u(c, q, a)$$

- ▶ Utility is maximized subject to the **budget constraint** $y = c + pq$,
 - ▶ where the wage y is the sole source of income, the unit rent of housing is p and the commodity price is the numeraire (normalized to one)
- ▶ Using the first-order conditions, we obtain the **indirect utility**

$$V = V(y, p, a)$$

- ▶ Naturally, we have $\frac{\partial V}{\partial y} > 0$, $\frac{\partial V}{\partial p} < 0$, $\frac{\partial V}{\partial a} > 0$
 - ▶ We view a as quality of life (a composite amenity)
 - ▶ a could also be a disamenity, in which case, of course, $\frac{\partial V}{\partial a} < 0$

Spatial equilibrium

- ▶ In equilibrium, workers must be **indifferent across locations**
 - ▶ If not, workers would **move to locations offering higher utility**
 - ▶ This will lead to **bidding up housing prices** or **pushing down wages**
 - ▶ The process will only stop when utilities are equalized everywhere
- ▶ Formally, this implies that the following condition must hold for all cities

$$V(y, p, a) = \bar{u}$$

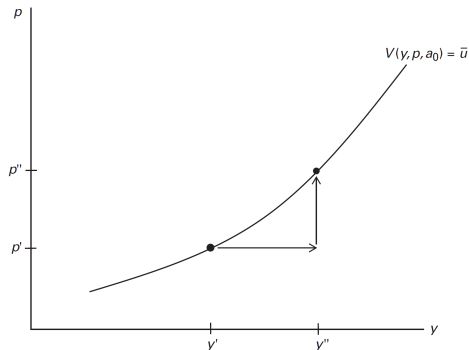
- ▶ For a given a , the indifference condition will only be satisfied if the effects of **differences in p and w across cities offset each other**

Spatial equilibrium is a core concept of spatial economics

Indifference curves are upward sloping

- ▶ Consider the point (p', y')
 - ▶ The bottom-left point
- ▶ Suppose that wage increases from y' to y''
 - ▶ Moving to the right
 - ▶ This change would raise utility
- ▶ Rent must increase
 - ▶ to keep utility constant
 - ▶ Moving up from p' to p''

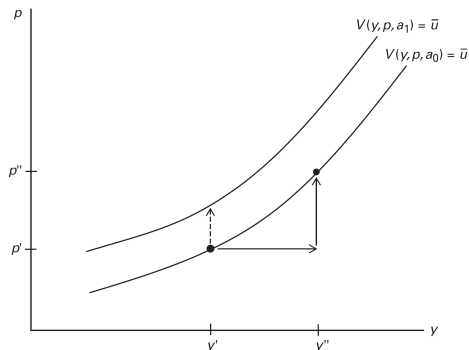
Q: How does the indifference curve shift if a increases?



Indifference curves with different amenity levels

- ▶ Now consider another city with amenity level $a_1 > a_0$
- ▶ Ceteris paribus, V would increase since $\frac{\partial V}{\partial a} > 0$
- ▶ To ensure $V = \bar{u}$ in both cities, we require that:
 - ▶ $p_1 > p_0$
 - ▶ $y_1 < y_0$
 - ▶ or a combination of both

Compensating differential



Firms

- ▶ Firms produce the non-housing commodity c using the **inputs**
 - ▶ **Labour**, hired at wage y
 - ▶ **Floor space** (of factories and offices), rented at rent p
 - ▶ One integrated floor space market, no land use frictions
 - ▶ Other inputs can be ignored, as long as their price is the same in all regions (capital)
- ▶ Constant returns to scale
 - ▶ Cost per unit of output is the same no matter how much is being produced
- ▶ **Cost function** $C = C(y, p, a)$ also **depends on** a , with $\frac{\partial C}{\partial y} > 0$, $\frac{\partial C}{\partial p} > 0$, $\frac{\partial C}{\partial a} \leq 0$
 - ▶ An amenity to workers may be an amenity to firms (low crime)
 - ▶ May be a disamenity to firms (regulation of pollution)

Spatial equilibrium (again)

- ▶ **Perfect competition and mobility** ensure that firms make **zero profits**
 - ▶ If a city offers greater profits it will attract firms
 - ▶ This will increase factor prices (real estate, labor)
 - ▶ Until **zero-profit equilibrium** is restored
- ▶ Since the price of c is the numeraire ($=1$), the **spatial equilibrium condition** is:

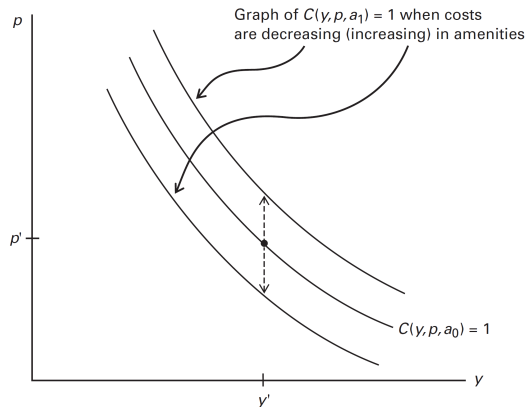
$$C(y, p, a) = 1$$

- ▶ Just saying that the unit cost C must equal the price of c
- ▶ This condition generates **iso-profit curves** in the p and y space
 - ▶ Combinations of p and y that yield zero profits for a given level of a

Does a shift iso-profit curve up or down?

Iso-profit Curves

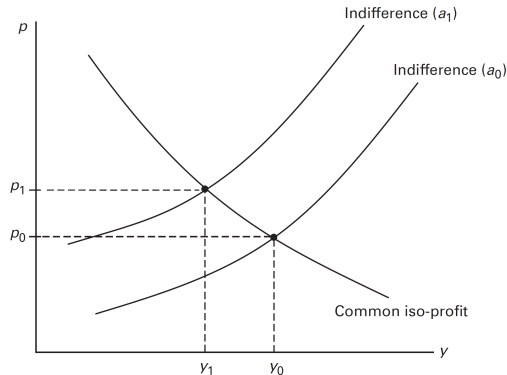
- ▶ Iso-profit curves are downward-sloping
 - ▶ For a given a , a higher y must be compensated by a lower p to keep unit cost constant
- ▶ If the amenity lowers costs, $a_1 > a_0$:
 - ▶ $p_1 > p_0$
 - ▶ $y_1 > y_0$
 - ▶ or a combination of both
- ▶ If the amenity increases costs
 - ▶ compensating differentials will point in opposite direction



Comparative statics: No amenity effect on firm cost

- ▶ **Equilibrium** must satisfy
 - ▶ indifference condition
 - ▶ iso-profit condition
- ▶ Holds only for the **intersection point**
 - ▶ Gives market-clearing p and y
- ▶ $a_1 > a_0$, shifts only indifference curve
 - ▶ No effect on firm costs
- ▶ Move uphill along the iso-profit curve

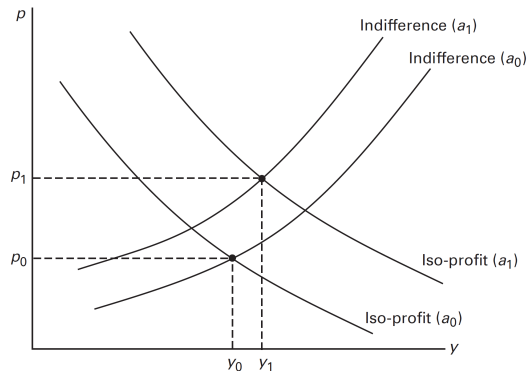
Workers accept **lower wage** and **higher rent** in more attractive city



Comparative statics: Producer amenity and consumer amenity

- ▶ $a_1 > a_0$ shifts iso-profit curve up
 - ▶ Firms can afford higher $\{y, p\}$
- ▶ Two forces pushing for higher rents
 - ▶ Firms accept higher rents
 - ▶ Residents accept higher rents
- ▶ Countervailing forces on wages
 - ▶ Firms can afford higher wages
 - ▶ Residents accept lower wages

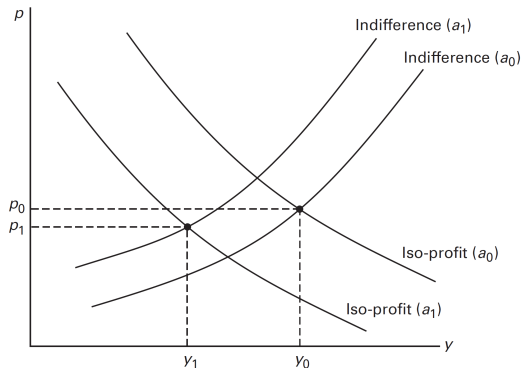
Nominal wage may be higher in more attractive city. But workers still accept **lower real wage**



Comparative statics: Producer disamenity and consumer amenity

- ▶ $a_1 > a_0$ shifts iso-profit curve down
 - ▶ Firms need to save on $\{y, p\}$
- ▶ Two forces pushing for lower wages
 - ▶ Firms need to save costs
 - ▶ Residents accept lower wages
- ▶ Countervailing forces on rents
 - ▶ Firms need to save costs
 - ▶ Residents accept higher rents

Rent may be lower in more attractive city. But workers still accept **lower real wage**



Equilibrium recap

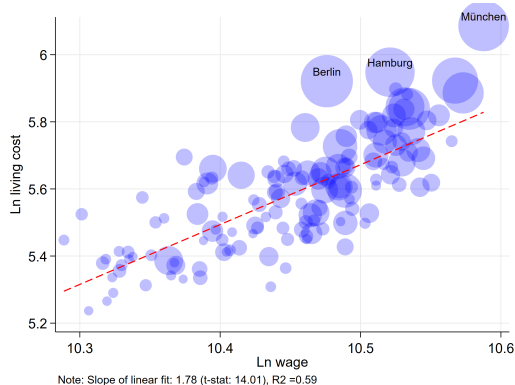
- ▶ City with **high quality of life** always has **low real wage**
 - ▶ City with high quality of life and **high production amenity** (low cost)
 - ▶ High wage and even higher rent
 - ▶ City with high quality of life and **low production amenity** (high cost)
 - ▶ High rent, but not necessarily high wage

What type of city is Berlin?

High or low quality of life? high or low production amenity?

Wages and rents in Germany

- ▶ Cities high above fitted line, have a high quality of life
 - ▶ See, Albouy and Lue (2015)
- ▶ Need to control for worker quality differences across regions
 - ▶ Wage index from between-city movers controlling for individual fixed effects
 - ▶ IAB data, 100% sample of employees
- ▶ Need to control for house quality differences across regions
 - ▶ Ahlfeldt, Heblich, Seidel (2022)



Quantitative framework

A quantitative framework

- ▶ Roback's (1982) theory is elegant and general
 - ▶ Avoids restrictions to specific functional forms
- ▶ For **quantitative analyses**, we need a parametric model
 - ▶ **Estimate** key parameters
 - ▶ **Rationalize** observed data (e.g. wages and prices) in levels and changes
 - ▶ **Conduct counterfactuals**—predict what happens if fundamentals change
- ▶ **Glaeser & Gottlieb (2009)** provide a quantitative spatial equilibrium framework
 - ▶ Very much in the tradition of Rosen-Roback
 - ▶ Competitive market in goods and factors
 - ▶ Free mobility and spatial equilibrium
 - ▶ Identical agents
 - ▶ Notice that cities are nested in a larger economy; no fixed worker endowment

A simple quantitative model following Glaeser & Gottlieb (2009)

- ▶ **Quantitative models** typically consist of
 - ▶ **Primitives** that are **exogenous**, including fundamentals, endowments, and parameters
 - ▶ **Endogenous objects** that are solved within the model
- ▶ Gottlieb and & Glaeser (2009) model cities, indexed by c , which differ in terms of

Exogenous Factors

- ▶ Productivities – A_c .
- ▶ Amenities – B_c .
- ▶ Land availability – M_c .

Endogenous Elements

- ▶ Population – L_c .
- ▶ Housing prices – P_c .
- ▶ Wages – W_c .

- ▶ Also a range of **parameters** which do not vary between cities

Labor Market I

- ▶ Competitive firm(s) produce a tradable good with $p_q = 1$ (Cobb-Douglas)

$$Q = A_c L^\alpha K^\beta \bar{Z}^{1-\alpha-\beta}$$

- ▶ L is the labor factor, priced per city at w_c
 - ▶ K is tradable capital, common price $r = 1$.
 - ▶ $\bar{Z}$ is a fixed factor in the city (e.g., land allocated for production).
- ▶ Maximize profit:

$$\max_{L,K} Q(L, K, \bar{Z}) - wL - K$$

- ▶ **First-Order Conditions**

- ▶ for K : $\frac{\partial \pi}{\partial K} = \beta A_c L^\alpha \bar{Z}^{1-\alpha-\beta} = K^{1-\beta}$
 - ▶ for L : $\frac{\partial \pi}{\partial L} = \alpha A_c K^\beta \bar{Z}^{1-\alpha-\beta} = L^{1-\alpha} w_c$

Labor Market II

► Inverse demand function:

- Substituting K from FOC for capital into FOC for labor condition
- taking logs

$$\log(w_c) = \Omega_1 + \frac{\log(A_c)}{1 - \beta} - \frac{1 - \alpha - \beta}{1 - \beta} \log(L_c) + \frac{1 - \alpha - \beta}{1 - \beta} \log(\bar{Z})$$

- wage w_c by fundamental productivity
- w_c falls in population L_c
 - Downward slope generated by the fixed factor $\bar{Z}$
- Notice that unlike in Roback (1982) **floor space is not an input** into production

Housing Market

- ▶ **Housing** demand generated by the resident population
 - ▶ Preferences $u(x, h) = B_c x^{1-\sigma} h^\sigma$
 - ▶ Use F.O.C. to derive Marshallian demand $h = \frac{\sigma w_c}{P_c}$
 - ▶ Housing demand: $H_D = h_c L_c = \frac{\sigma w_c}{P_c} L_c$
- ▶ **Supply of housing** produced by developers using one unit of land
 - ▶ Convex construction costs: $c(D) = dD^\delta$, where D is the building height & $\delta > 1$
 - ▶ Abstracting from land costs, max. profits $\Pi = (P_c D - \delta_0 D^\delta)$ yields $D_c^* = \left(\frac{P_c}{d_0 \delta}\right)^{\frac{1}{\delta-1}}$
 - ▶ Housing supply: $H_S = M_c \times D_c^* = M_c \left(\frac{P_c}{d_0 \delta}\right)^{\frac{1}{\delta-1}}$
- ▶ Housing market clears in **equilibrium**: $H_D = H_S$

$$\Rightarrow \log(P_c) = \Omega_2 + \frac{\delta - 1}{\delta} [\log(w_c) + \log(L_c) - \log(M_c)]$$

Spatial equilibrium

- ▶ Cities have the same level of utility
 - ▶ Just like in Roback (1982)
- ▶ Max utility: $u(x, h) = B_c x^{1-\sigma} h^\sigma$ s. t. budget constraint: $P_c h_c + x = W_c$
 - ▶ Indirect utility: $V(W_c, P_c, B_c) = (1 - \sigma)^{1-\sigma} \sigma^\sigma \frac{B_c W_c}{P_c^\sigma}$
 - ▶ Notice that $p_q = 1$ (see Labour Market I)
- ▶ **Equilibrium**
 - ▶ Constant utility in urban system: $V(W_c, P_c, B_c) = \bar{V}$
- ▶ **Indifference condition** (in logs):
$$\log(W_c) = \Omega_3 - \log(B_c) + \sigma \log(P_c)$$
 - ▶ Upward-sloping just like in Roback (1982)

General equilibrium I

- ▶ We have derived a system of **three** equations (using two equilibrium conditions):

$$\log(w_c) = \Omega_1 + \frac{\log(A_c)}{1 - \beta} - \frac{1 - \alpha - \beta}{1 - \beta} \log(L_c) + \frac{1 - \alpha - \beta}{1 - \beta} \log(Z)$$

$$\log(P_c) = \Omega_2 + \frac{\delta - 1}{\delta} [\log(w_c) + \log(L_c) - \log(M_c)]$$

$$\log(w_c) = \Omega_3 - \log(B_c) + \sigma \log(P_c)$$

- ▶ Has **three** endogenous objects: Population (L_c), wages (w_c), housing prices (P_c)
- ▶ All the other objects are exogenous: (A_c), (M_c), (B_c), and parameters

As many equations as unknowns $\Rightarrow$ Solve for the general equilibrium ✓

General equilibrium II

- Solve for the endogenous objects and express them in terms of exogenous objects

$$\log(L_c) = \Omega_4 + \eta(\delta + \sigma - \sigma\delta) \log(\hat{A}_c) + \eta(1 - \beta)(\delta \log(B_c) + \sigma(\delta - 1) \log(M_c))$$

$$\log(w_c) = \Omega_5 + \sigma\eta(\delta - 1) \log(\hat{A}_c) - \eta(1 - \alpha - \beta)(\delta \log(B_c) + \sigma(\delta - 1) \log(M_c))$$

$$\log(P_c) = \Omega_6 + (\delta - 1)\eta(\log(\hat{A}_c) + \alpha \log(B_c) - (1 - \alpha - \beta) \log(M_c))$$

- with $\eta = [(1 - \beta)\delta - \alpha(\sigma + \delta - \sigma\delta)]^{-1}$, $\log(\hat{A}_c) = \log(A_c) + (1 - \alpha - \beta) \log(Z_c)$

► **Useful homework:** Rearrange eqs. from previous slide to obtain above eqs.!

- Can now solve for endogenous objects for any given values of primitives

Ready for counterfactuals (comperative statics) ✓

Counterfactuals

- ▶ Can **quantify the effect of a change in an exogenous location characteristic**:

$$\ln Y_c^1 = \frac{\partial \ln Y_c}{\partial \ln X_c} d \ln X_c + \ln Y_c^0$$

- ▶ where superscript 1 denotes the counterfactual and 0 denotes the initial value
- ▶ **Play around with the GG2009-toolkit to strengthen your intuition!**
 - ▶ Will deliver $\Delta \ln Y$ for user-specified $\Delta \ln X$ [▶ Open GG2009-toolkit](#)
- ▶ Qualitatively, we obtain the following derivatives

	ΔA_c	ΔB_c	ΔM_c
ΔL_c	+	+	+
Δw_c	+	-	-
ΔP_c	+	+	-

Estimation

- ▶ Need to **quantify the model** before we can work quantitatively
 - ▶ First step is to settle on **parameter values**, which we can **set or estimate**
- ▶ For example, solve the inverse labour supply equation for $\log(L_c)$

$$\log(L_c) = \underbrace{\Omega_1 \times \frac{1 - \beta}{1 - \alpha - \beta} + \log(\bar{Z})}_{\text{constant}} \underbrace{- \frac{1 - \beta}{1 - \alpha - \beta}}_{\text{labour demand elasticity}} \log(w_c) + \underbrace{\frac{1}{1 - \alpha - \beta} \log(A_c)}_{\text{structural residual}}$$

- ▶ Can estimate using labour demand elasticity in **reduced-form**
 - ▶ Can infer β if we know α and vice versa
 - ▶ Identifying variation in $\log(w_c)$ must be uncorrelated with demand shifter, A_c
 - ▶ **Excludable instrument** shifts labour supply, exclusively, e.g. B_c

Structural model helps developing transparent identification strategy

Inversion

- ▶ To fully quantify the model, we also need to observe or **invert fundamentals**
 - ▶ Using observed values of endogenous variables and given parameter values
- ▶ To recover quality of life, B_c , we can rearrange the indifference condition.

$$\log(B_c) = \Omega_3 + \sigma \log(P_c) - \log(W_c)$$

- ▶ To identify B_c up to a multiplicative constant we just need:
 - ▶ Need wages, w_c , rents (or prices) P_c and the expenditure share of housing, $\sigma = 0.33$
 - ▶ Typical value used in the literature (see Ahlfeldt & Pietrostefani, 2019)

Have fun with the inversion in the tutorial

Valuing amenities

Valuing non-marketed goods

- ▶ Recovering B_c is not only useful to measure quality of life
 - ▶ And compute quality-of-life rankings
- ▶ We use the 'compensating differential' to value non-marketed goods

$$\ln B_c = \sum_j \gamma^j \ln X_c^j + \varepsilon_c,$$

- ▶ where X_c^j is one of $j \in J$ (dis)amenities
- ▶ It is straightforward (in principle) to estimate 'hedonic' price $\gamma^j = \frac{\partial \ln B_c}{\partial \ln X_c^j}$
 - ▶ Straightforward to monetize since $\frac{\partial \ln V}{\partial \ln w_c} = \frac{\partial \ln V}{\partial \ln B_c}$

This how we value clean air, crime, public schools... (Greenstone, 2017)

Conclusion

- ▶ **Rosen-Roback framework is a core concept in spatial economics**
 - ▶ A family of models that share the idea of a frictionless spatial equilibrium
 - ▶ Perfect mobility implies indifference, zero-profits and perfect spatial arbitrage
- ▶ Framework can be used to **recover quality of life**
 - ▶ From spatial variation in wages and rents
- ▶ Workhorse tool for **'revealed-preference' valuation non-marketed goods**
 - ▶ Very important for social cost-benefit analysis and public investment decisions

Next week: **Monocentric city model**

Literature

Core readings

- ▶ Brueckner, J. (2011): **Wage-Based Indexes of Urban Quality of Life. Lectures on urban economics. Chapter 11. MIT Press.**
- ▶ Glaeser, E., J. Gottlieb (2009): **The Wealth of Cities: Agglomeration Economies and Spatial Equilibrium in the United States. Journal of Economic Literature 2009, 47(4).**

Other readings

- ▶ Ahlfeldt, G., Pietrostefani, E. (2019): The economic effects of the density: A synthesis. *Journal of Urban Economics*, 111.
- ▶ Ahlfeldt, G., S. Heblich, T. Seidel (2023): Micro-geographic property price and rent indices. *Regional Science and Urban Economics*, 98.
- ▶ Albouy, D. and B. Lue (2015): Driving to Opportunity: Local Rents, Wages, Commuting, and Sub-Metropolitan Quality-of-Life. *Journal of Urban Economics*, September 2015, 89, 74-92.
- ▶ Greenstone, M. (2017): The Continuing Impact of Sherwin Rosen's "Hedonic Prices and Implicit Markets: Product Differentiation in Pure Competition". *Journal of Political Economy*, 126(6).
- ▶ Rosen, S. (1979). Wage-Based Indexes of Urban Quality of Life. In *Current Issues in Urban Economics*, edited by P. Mieszkowski and M. Straszheim. Baltimore: Johns Hopkins University Press.
- ▶ Roback, J. (1982). Wages, Rents, and the Quality of Life. *Journal of Political Economy* 90(4), 1257-1278.